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Aligning Orphanet Classification to Identify Disease Characteristics among Rare Disease Clusters

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Abstract

Understanding the underlying etiologies of rare diseases may facilitate research across multiple conditions, enabling basket trial design and drug repurposing. In this study, we aligned clusters of rare diseases with Orphanet classifications to represent their shared etiologies and establish a foundation for further investigation on underlying biological mechanism discovery. By utilizing the linearized Orphanet categories, we connected 35 clusters of rare diseases into 18 classifications. Significant associations were found between the categories “Rare Developmental Defects During Embryogenesis” and “Rare Inborn Errors of Metabolism” and the clusters in this study, suggesting that many rare diseases originating in the prenatal period or related to metabolism may present a substantial opportunity for success in future investigation.

Keywords

rare disease; disease cluster; Orphanet classification; rare disease characteristics

I. Introduction

Rare diseases impact a limited number of individuals, explicitly affecting fewer than 200,000 individuals in the United States or fewer than 1 in 2,000 individuals in Europe [1,2]. Due to the wide variety of disease presentations, rare diseases often involve lengthy diagnostic processes and may go undiagnosed or misdiagnosed, presenting significant

challenges [3,4]. Furthermore, many rare diseases have few effective treatment options and incur high medical costs due to associated complications [5,6]. To tackle those challenges, various research and treatment strategies for multiple rare diseases simultaneously, such as basket clinical trials or drug repurposing, can be effective, specifically shared underlying causal factors of targeting rare diseases [7].

Our preliminary study demonstrated that clusters of rare diseases can be formed based on their underlying causal genes and pathophysiology [8]. These clusters were created using graph node embeddings derived from a knowledge graph that analyzed the overlap between rare diseases tracked by the Genetic and Rare Diseases Information Center (GARD) [9] and Orphanet [10], enriched with additional information on small molecules and biological pathways. However, the clusters produced from this study are associated with broad scopes. For instance, some clusters encompass more than 100 diseases, which has limitations in etiologies.

The Orphanet classification is a multi-hierarchical system for classifying rare diseases based on clinical criteria and medical specialties [10]. It considers factors such as clinical features, etiology, affected systems, and the genetic basis of the diseases [11]. In this study, we systematically aligned clusters of rare diseases with the Orphanet classification to reveal their underlying etiologies within or between the cluster and provide a way to utilize shared mechanisms in future approaches such as basket clinical trials or drug repurposing.

II. Method

A. Dataset

Our previous study resulted in 37 distinct clusters of rare diseases, with each cluster averaging 87 diseases (mean 87.62, standard deviation 30.23, and median 83), totaling 3,242 diseases. Each rare disease is associated with a GARD ID, and a disease name represented as a string.

We used three Orphanet classification systems. The first is the Orphanet Rare Disease Ontology (ORDO), which provides a structured vocabulary for rare diseases [12]. This ontology includes the classification of rare diseases (ORPHA code), relationships between diseases and genes, and connections to other terminologies and databases in OWL format (ORDO_en_4.3.owl). The second, “Rare Diseases and Alignment with Terminologies and Databases,” provides references for various terminologies (en_product1_new.xml) [13]. The third one is the Orphanet linearization (en_product7.xml) [14]. This linearization classification is designed to manage rare diseases from a mono-hierarchical perspective, where each disease is assigned to a single medical specialty. Linearization parents are selected from the classification’s top entities (n=31) [15]. We connected rare diseases to Orphanet linearization classifications to simplify the complex representation of their underlying etiologies.

B. ORPHA code Identification for GARD rare diseases

Two steps were employed to identify the ORPHA code for each rare disease: reference matching and string matching. We utilized reference information associated with the GARD

ID, ORPHA code, and other relevant terminology codes (such as ICD, UMLS, OMIM, etc.) by searching the linked data in the <reference> notation within the “Rare Diseases and Alignment with Terminologies and Databases” file. If the reference matching failed, we proceeded with string matching. The string matching involved searching for the GARD disease name in the <label> and <alternative term> notations in the ORDO XML file for the corresponding ORPHA code.

C. Linearization classification mapping

We assigned the linearized class to each GARD rare disease based on the mapped ORPHA codes from the above step.

D. Enrichment analysis

We performed Fisher’s exact test [16] for each cluster, comparing the counts of linearized classifications within that cluster against those in the other clusters. Fisher’s exact test was used to assess non-random associations between categorical variables in small sample sizes. TABLE I is a contingency table used for Fisher’s exact test. Fisher’s exact test results were adjusted for multiple comparisons using the Benjamini-Hochberg procedure, with an alpha level of 0.01, applying the false discovery rate method to identify significant associations while limiting the proportion of false positives. The enrichment results are visualized as Sankey diagrams, illustrating the relationships between diseases in the clusters and orphaned linearized classifications using the Plotly library in Python [17].

III. Result

The disease mapping results between GARD and Orphanet were successful for 3,242 out of 3,260 rare diseases, representing a 99.44% match.

By enriching linearized classifications, we were able to map 35 out of 37 clusters to 18 Orphanet linearized classes. As shown in Fig. 1, the majority of these clusters were enriched to two specific linearized classifications, “Rare Developmental Defects During Embryogenesis (ORPHA93890)” with 10 clusters and “Rare Inborn Errors of Metabolism (ORPHA68367)” with 7 clusters.

A single cluster may correspond to one or multiple linearized Orphanet classifications. For instance, the diseases in Cluster 17 encompass all relevant visual organs and associated eyes, leading to its sole linearized classification as a “Rare Ophthalmic Disease (ORPHA97966)” exclusively. In the case of Cluster 2, about half of the diseases corresponded to “Rare Systemic or Rheumatologic Diseases (ORPHA98023),” while the other half were linked to “Rare Infectious Diseases (ORPHA68416)”. This suggests that the rare diseases in Cluster 2 may exhibit a nuanced connection between rheumatologic diseases and infectious diseases like Infection-Induced Arthritis (e.g., Rheumatic Fever, GARD:0005699) or Autoimmune Responses in rheumatologic conditions (e.g., Adult-onset Still’s disease, GARD:0000436).

Among the unmapped linearized classifications in our clusters, some are related to surgical diseases, such as “Rare Surgical Cardiac Disease (ORPHA97965),” “Rare Surgical Thoracic Disease (ORPHA97962),” “Rare Abdominal Surgical Disease (ORPHA17578),” and “Rare

Maxillo-Facial Surgical Disease (ORPHA17578). These may be associated with clinical events rather than the genetic or pathway information that initially generated the clusters of rare diseases.

IV. Discussion

This study identified the underlying etiology of rare disease clusters, as enriched by Orphanet linearized classifications. By focusing on 18 Orphanet linearized classifications, we observed general characteristics of these clusters, such as their reliance on genetic information rather than clinical information for classification. Specifically, we identified “Rare Developmental Defects During Embryogenesis (ORPHA93890)” and “Rare Inborn Errors of Metabolism (ORPHA68367)” as the primary underlying etiologies for these clusters.

The Orphanet linearized classification describes rare diseases only limitedly, as it tends to select one parent classification from a specific set of entities in the hierarchy and focuses primarily on the organ itself [15]. Specifically, developmental anomalies are prioritized in linearized categories over diseases affecting the body system. As a result, “Rare Developmental Defects During Embryogenesis (ORPHA93890)” includes many rare diseases in our study.

Additionally, linearized classification simplifies the representation by reducing complexity. When a disease affects only one specific organ, it is classified solely under that organ. Consequently, the “Inborn Errors of Metabolism (ORPHA68367)” classification often addresses multiple organ issues instead of encompassing all organs, excluding the problem that impacts just one organ. In other words, some diseases outside our 10 Clusters, which respond solely to one organ, may be relevant to “Inborn Errors of Metabolism (ORPHA68367)” but were omitted from our Fig. 1.

We also review the diseases within the clusters aligned with “Rare Developmental Defects During Embryogenesis (ORPHA93890)” in Fig. 1, as each disease is linked to a specific Orphanet category, identified by its unique ORPHA code. These clusters may be categorized depending on the anatomical locations or dysfunction of mechanisms. For example, diseases in Clusters 32, 34, and 26 are highly associated with intellectual disabilities; Clusters 21 and 6 pertain to bone diseases; Cluster 27 relates to skin disorders, and Cluster 26 is associated with eye diseases. This indicates that the Orphanet originated the affected systems classifications, which can illustrate different body locations based on the cluster despite these clusters sharing the exact etiology or underlying mechanism.

We aim to expand our study to explore the specific mechanisms shared within the cluster, using the case study of “Rare Inborn Errors of Metabolism (ORPHA68367),” which serves as the primary focus of our research.

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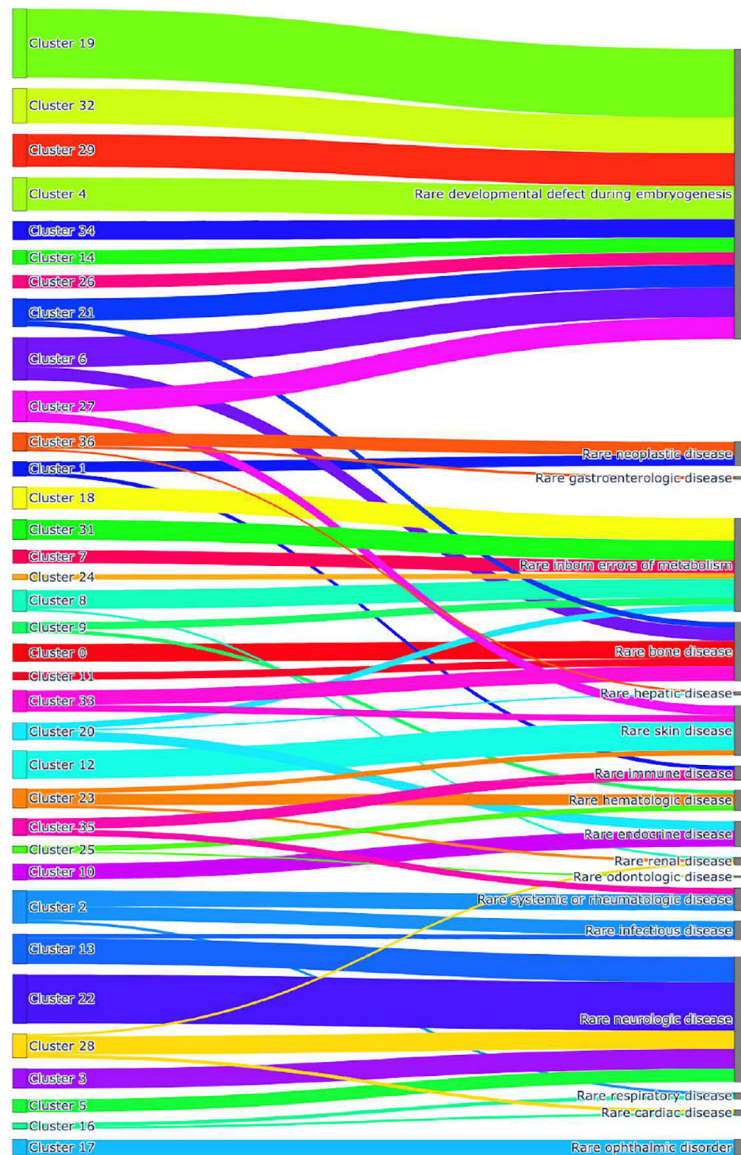


Fig. 1.
Rare Disease Clusters to the Orphanet Linearization Category

TABLE I.

A Contingency Table for Fisher's exact test

	Orphanet linearized classification j	Not in Orphanet linearized classification j
In Cluster i	A^a	B^b
Not in Cluster i	C^c	D^d

^aThe count of rare diseases in cluster i that are presenting in Orphanet linearized classification j

^bThe count of rare diseases in cluster i that are not presenting in Orphanet linearized classification j

^cThe count of rare diseases not in cluster i that are presenting in Orphanet linearized classification j

^dThe count of rare diseases not in cluster i that are not presenting in Orphanet linearized classification j